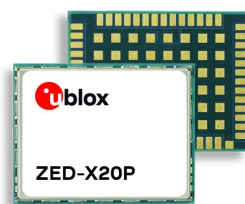


ZED-X20P-00B

All-band high precision GNSS module

Professional grade

Data sheet



Abstract

This data sheet describes the ZED-X20P high precision module with all-band GNSS receiver. The module provides all-band RTK with fast convergence times, reliable performance and easy integration of RTK for fast time-to-market. It has a high update rate for highly dynamic applications and centimeter-level accuracy in a small and energy-efficient module.

Document information

Title	ZED-X20P-00B	
Subtitle	All-band high precision GNSS module	
Document type	Data sheet	
Document number	UBXDOC-963802114-12690	
Revision and date	R04	22-Sept-2025
Disclosure restriction	C1-Public	

Product status	Corresponding content status	
Functional sample	Draft	For functional testing. Revised and supplementary data will be published later.
In development / prototype	Objective specification	Target values. Revised and supplementary data will be published later.
Engineering sample	Advance information	Data based on early testing. Revised and supplementary data will be published later.
Initial production	Early production information	Data from product verification. Revised and supplementary data may be published later.
Mass production / End of life	Production information	Document contains the final product specification.

This document applies to the following products:

Product name	Type number	FW version	IN/PCN reference	Product status
ZED-X20P	ZED-X20P-00B-00	HPG 2.02	N/A	Initial production

u-blox or third parties may hold intellectual property rights in the products, names, logos and designs included in this document. Copying, reproduction, or modification of this document or any part thereof is only permitted with the express written permission of u-blox. Disclosure to third parties is permitted for clearly public documents only.

The information contained herein is provided "as is" and u-blox assumes no liability for its use. No warranty, either express or implied, is given, including but not limited to, with respect to the accuracy, correctness, reliability and fitness for a particular purpose of the information. This document may be revised by u-blox at any time without notice. For the most recent documents and product statuses, visit www.u-blox.com.

Copyright © 2025, u-blox AG.

Contents

Document information.....	2
Contents.....	3
1 Functional description.....	4
1.1 Overview.....	4
1.2 Performance.....	4
1.3 Supported GNSS constellations.....	5
1.4 Supported assisted GNSS (A-GNSS) services.....	6
1.5 Supported augmentation systems.....	6
1.6 Differential GNSS (DGNSS).....	7
1.7 Supported protocols.....	8
1.8 Firmware features.....	8
1.9 Broadcast navigation data and satellite signal measurements.....	8
1.9.1 Carrier-phase measurements.....	9
2 Block diagram.....	10
3 Pin definition.....	11
3.1 Pin assignment.....	11
3.2 Pin state.....	13
4 Electrical specifications.....	14
4.1 Absolute maximum ratings.....	14
4.2 Operating conditions.....	14
4.3 Indicative power requirements.....	15
5 Communications interfaces.....	16
5.1 UART.....	16
5.2 SPI.....	16
5.3 I2C.....	18
5.4 Default interface settings.....	19
6 Mechanical specifications.....	21
7 Qualifications and approvals.....	23
8 Packaging.....	24
8.1 Reels.....	24
8.2 Tapes.....	24
9 Soldering.....	25
10 Product marking and ordering information.....	26
10.1 Product marking.....	26
10.2 Product identifiers.....	26
10.3 Ordering codes.....	27
Related documents.....	28
Revision history.....	29
Contact.....	30

1 Functional description

1.1 Overview

ZED-X20P is an innovative all-band receiver module designed to revolutionize positioning technology in industrial applications. Built upon the u-blox new generation receiver platform, this module offers all-band Global Navigation Satellite System (GNSS) capability, supporting bands including L1, L2, L5 and L6. With its comprehensive coverage, ZED-X20P ensures precise and reliable positioning even in challenging environments, setting a new standard in accuracy.

Equipped with integrated u-blox allband real-time kinematic (RTK) and precise point positioning real-time kinematic (PPP-RTK) technologies, ZED-X20P achieves centimeter-level accuracy, enabling precise navigation and automation in industrial and consumer-grade products. Despite its advanced capabilities, ZED-X20P maintains a compact surface-mounted form factor, measuring only 17.0 x 22.0 x 2.4 mm, ensuring seamless integration into various applications without compromising performance.

In this document, RTK refers to an observation state representation (OSR) based solution utilizing radio technical commission for maritime services (RTCM) corrections, while PPP-RTK refers to state space representation (SSR) based solution using secure position augmentation for real-time navigation (SPARTN). With its comprehensive features and advanced technologies, ZED-X20P offers unparalleled accuracy and reliability, making it the ideal choice for applications requiring high-performance positioning solutions.

1.2 Performance

Parameter	Specification	Value
Receiver type		All-band high precision GNSS receiver
Accuracy of time pulse signal	RMS	20 ns
	99%	60 ns
Frequency of time pulse signal		Default 1PPS (0.25 Hz to 10 MHz configurable)
Operational limits ¹	Dynamics	≤ 4 g
	Altitude	80,000 m
	Velocity	300 m/s
Velocity accuracy ²		0.03 m/s
Dynamic heading accuracy ²		0.3 deg

Table 1: ZED-X20P specifications

GNSS		GPS+GAL+BDS
Acquisition ³	Cold start	25 s
	Hot start	2 s
	Aided start ⁴	2 s

¹ Assuming Airborne 4 g platform.

² 50% in the range of 5 to 30 m/s for dynamic operation.

³ Commanded starts. All satellites at -130 dBm. Measured at room temperature. All-band operation

⁴ Depends on the speed and latency of the aiding data connection, commanded starts

GNSS		GPS+GAL+BDS
Max navigation update rate ⁵	RTK	25 Hz
	PVT	25 Hz
	RAW	25 Hz
Convergence time ⁶	RTK	< 7 s

Table 2: ZED-X20P performance

GNSS		GPS+GAL+BDS
Horizontal position accuracy (CEP)	PVT ⁷	1.2 m
	SBAS ⁷	0.6 m
	RTK ⁸	0.006 m
		+ 1 ppm
Vertical position accuracy (Median)	PVT ⁷	2.0 m
	SBAS ⁷	1.0 m
	RTK ⁸	0.01 m
		+ 1 ppm

Table 3: ZED-X20P position accuracy

GNSS		GPS+GAL+BDS
Horizontal position accuracy (CEP)	SPARTN ⁹	< 0.06 m
Vertical position accuracy (Median)	SPARTN ⁹	< 0.10 m
Convergence time ⁶	SPARTN ⁹	< 40 s

Table 4: ZED-X20P performance for PPP-RTK

GNSS		GPS+GAL+BDS
Sensitivity ¹⁰	Tracking and nav.	-167 dBm
	Reacquisition	-160 dBm
	Cold start	-148 dBm
	Hot start	-158 dBm

Table 5: ZED-X20P sensitivity

1.3 Supported GNSS constellations

ZED-X20P is a concurrent GNSS receiver capable of simultaneously receiving and tracking signals from multiple global navigation satellite constellations. Its advanced RF front-end architecture enables concurrent reception of all major and regional GNSS systems as presented in [Table 6](#). Signals are received across all supported bands, ensuring enhanced positioning accuracy and reliability.

⁵ Measured with primary output only, secondary output disabled (default)

⁶ Depends on atmospheric conditions, baseline length, GNSS antenna, multipath conditions, satellite visibility and geometry

⁷ 24-hours static measurement

⁸ Accuracy is based on empirical testing over a 1 km baseline using patch antennas mounted on stable ground planes.

Antenna phase center offset errors are excluded. A +1 ppm scale error typically contributes ~1 mm of error per km, applicable for baselines up to 20 km. At 20 km, the total positioning error is approximately 3 cm. These values are based on observed performance and are not guaranteed design limits.

⁹ PointPerfect Flex correction service based on the SPARTN format

¹⁰ Demonstrated with a good external LNA. Measured at room temperature.

All satellites in view can be processed to provide an RTK navigation solution when used with correction data.

System	Signal	Frequency
GPS	L1C/A	1575.420 MHz
	L2C	1227.600 MHz
	L5	1176.450 MHz
Galileo	E1-B/C	1575.420 MHz
	E5a	1176.450 MHz
	E6	1278.750 MHz
BeiDou	B1I	1561.098 MHz
	B1C	1575.420 MHz
	B2a	1176.450 MHz
	B3I	1268.520 MHz
QZSS	L1C/A, L1C/B ¹¹	1575.420 MHz
	L2C	1227.600 MHz
	L5	1176.450 MHz
NavIC	L1-SPS Data ¹¹	1575.420 MHz
	L5-SPS	1176.450 MHz

Table 6: Supported GNSS constellations and signals on ZED-X20P



If power consumption is a key factor, the receiver can be configured for a subset of GNSS constellations.

1.4 Supported assisted GNSS (A-GNSS) services

ZED-X20P supports the GNSS assistance services in [Table 7](#).

Service	Signal
AssistNow™ Online	GPS L1C/A
	Galileo E1
	QZSS L1C/A
	BeiDou B1I

Table 7: Supported assisted GNSS (A-GNSS) services

1.5 Supported augmentation systems

ZED-X20P supports the augmentation systems in [Table 8](#).

System	Signal
SBAS	EGNOS
	GAGAN
	MSAS
	WAAS
	BDSBAS ^{11, 12}

Table 8: Supported augmentation systems



The SBAS augmentation system can be enabled only if GPS operation is also enabled.

¹¹ Currently in development phase.

¹² BDSBAS is not operational and is in the test phase.

1.6 Differential GNSS (DGNSS)

When operating in Real time kinematic (RTK) mode, RTCM version 3 messages are required and the module supports DGNSS according to RTCM 10403.4.

Operating as a rover, ZED-X20P can decode the following RTCM 3.4 messages:

Message type	Description
RTCM 1001	L1-only GPS RTK observables
RTCM 1002	Extended L1-only GPS RTK observables
RTCM 1003	L1/L2 GPS RTK observables
RTCM 1004	Extended L1/L2 GPS RTK observables
RTCM 1005	Stationary RTK reference station ARP
RTCM 1006	Stationary RTK reference station ARP with antenna height
RTCM 1007	Antenna descriptor
RTCM 1033	Receiver and antenna description
RTCM 1074	GPS MSM4
RTCM 1075	GPS MSM5
RTCM 1077	GPS MSM7
RTCM 1094	Galileo MSM4
RTCM 1095	Galileo MSM5
RTCM 1097	Galileo MSM7
RTCM 1124	BeiDou MSM4
RTCM 1125	BeiDou MSM5
RTCM 1127	BeiDou MSM7

Table 9: Supported RTCM 3.4 input messages

Operating as a base station, ZED-X20P can generate the following RTCM 3.4 output messages:

Message type	Description
RTCM 1005	Stationary RTK reference station ARP
RTCM 1074	GPS MSM4
RTCM 1077	GPS MSM7
RTCM 1094	Galileo MSM4
RTCM 1097	Galileo MSM7
RTCM 1124	BeiDou MSM4
RTCM 1127	BeiDou MSM7

Table 10: Supported RTCM 3.4 output messages

Operating as a rover, ZED-X20P can decode the following SPARTN 2.0.2 messages:

Message type-subtype	Description
SM 0-0	GPS orbit, clock, bias (OCB)
SM 0-2	Galileo orbit, clock, bias (OCB)
SM 0-3	BeiDou orbit, clock, bias (OCB)
SM 1-0	GPS high-precision atmosphere correction (HPAC)
SM 1-2	Galileo high-precision atmosphere correction (HPAC)
SM 1-3	BeiDou high-precision atmosphere correction (HPAC)

Message type-subtype	Description
SM 2-0	Geographic area definition (GAD)

Table 11: Supported SPARTN version 2.0.2 input messages

1.7 Supported protocols

ZED-X20P supports the protocols in [Table 12](#).

Protocol	Type
UBX	Input/output, binary, u-blox proprietary
NMEA 4.11 (default), 4.10 and 4.0	Input/output, ASCII
RTCM 3.4	Input/output, binary
SPARTN	Input, binary

Table 12: Supported protocols

For specification of the protocols, see the Interface description [\[2\]](#).

1.8 Firmware features

Feature	Description
Antenna supervisor ¹³	Antenna supervisor for active antenna control and short circuit detection
RAW data	Provides carrier and code phase measurements of tracked signals
Assisted GNSS	AssistNow Online
RTK support	High-precision positioning
Local base functionality	Base station functionality with RTCM output
Spectrum analyzer	Monitors RF environment for possible interference and anomalies
Geofencing	Supports location-restriction and anti-theft functionality
Monitoring and management functions	Extensive functions to monitor embedded system health such as I/O queues, pin status and correction status
Personalization	Individualized identity to access high value features and subscriptions
Backup modes	Hardware and software

Table 13: Firmware features

Feature	Description
Anti-jamming	RF interference and jamming detection and reporting
Anti-spoofing	Spoofing detection and reporting
Galileo OSNMA	Galileo Open Service Navigation Message Authentication
Configuration lockdown	Receiver configuration can be locked by command
Secure boot	Only signed firmware images are executed
Secure storage	Tamper-resistant secure storage with end-to-end security
Navigation data validation	According to the DHS <i>GPS Receiver Allow List Development Guide</i>

Table 14: Security features

1.9 Broadcast navigation data and satellite signal measurements

ZED-X20P uses UBX-RXM-SFRBX message to output all the GNSS broadcast data upon reception from tracked satellites. This includes all the supported GNSS signals as well as the QZSS and

¹³ External components required, some pins need to be reconfigured.

SBAS augmentation services. See the Interface description[\[2\]](#) for the UBX-RXM-SFRBX message specification. The receiver can provide satellite signal information in a form compatible with the Radio Resource LCS Protocol (RRLP).

1.9.1 Carrier-phase measurements

ZED-X20P provides raw carrier-phase data for all supported signals, along with pseudorange, Doppler and measurement quality information. The data contained in the UBX-RXM-RAWX message follows the conventions of a multi-GNSS RINEX 3 observation file. For the UBX-RXM-RAWX message specification, see Interface description[\[2\]](#).



Raw measurement data is available once the receiver has established data bit synchronization and time-of-week.

2 Block diagram

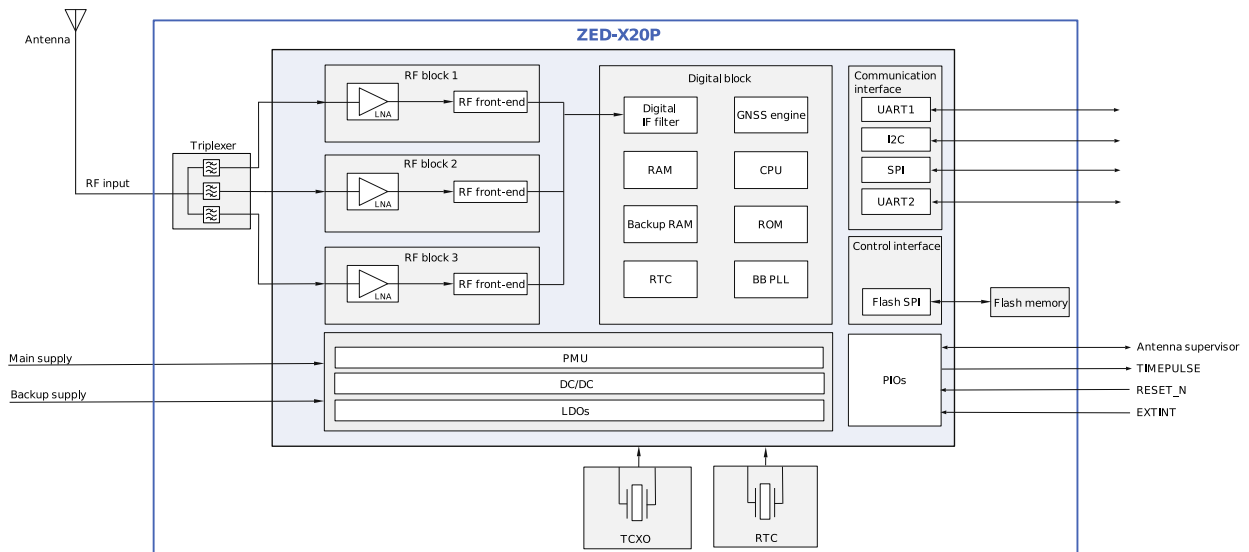


Figure 1: ZED-X20P block diagram

3 Pin definition

3.1 Pin assignment

The pin assignment of the ZED-X20P module is shown in [Figure 2](#). The defined configuration of the programmable input/outputs (PIOs) is listed in [Table 15](#).

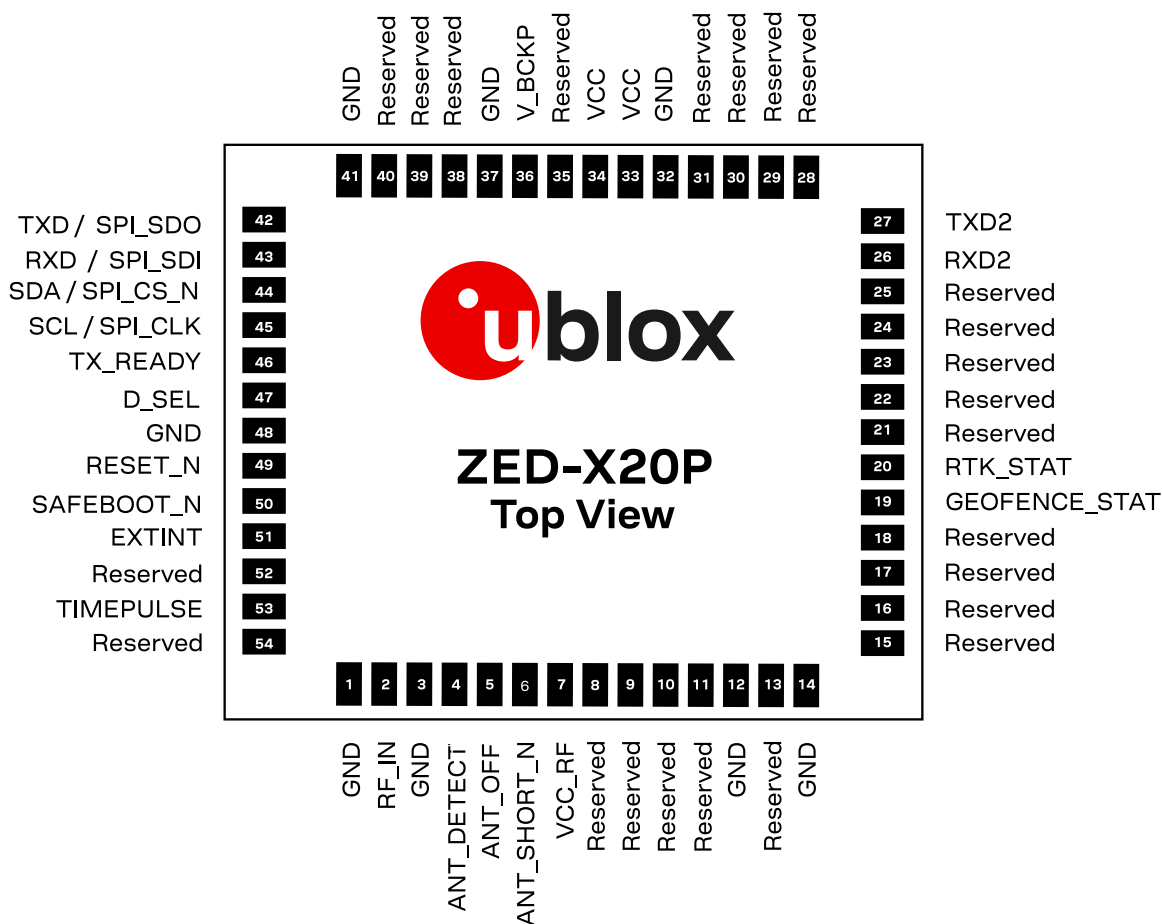


Figure 2: ZED-X20P pin assignment

Pin no.	Name	I/O	Description
1	GND	-	Ground
2	RF_IN	I	RF input
3	GND	-	Ground
4	ANT_DETECT	I	Active antenna detect - default active high
5	ANT_OFF	O	External LNA disable - default active high
6	ANT_SHORT_N	I	Active antenna short detect - default active low
7	VCC_RF	O	Voltage for external LNA
8	Reserved	-	Reserved
9	Reserved	-	Reserved
10	Reserved	-	Reserved

Pin no.	Name	I/O	Description
11	Reserved	-	Reserved
12	GND	-	Ground
13	Reserved	-	Reserved
14	GND	-	Ground
15	Reserved	-	Reserved
16	Reserved	-	Reserved
17	Reserved	-	Reserved
18	Reserved	-	Reserved
19	GEOFENCE_STAT	O	Geofence status, user defined
20	RTK_STAT	O	RTK status: 0 = RTK/PPP-RTK fixed Blinking = receiving and using corrections 1 = no corrections
21	Reserved	-	Reserved
22	Reserved	-	Reserved
23	Reserved	-	Reserved
24	Reserved	-	Reserved
25	Reserved	-	Reserved
26	RXD2	I	UART2 input
27	TXD2	O	UART2 output
28	Reserved	-	Reserved
29	Reserved	-	Reserved
30	Reserved	-	Reserved
31	Reserved	-	Reserved
32	GND	-	Ground
33	VCC	I	Voltage supply
34	VCC	I	Voltage supply
35	Reserved	-	Reserved
36	V_BCKP	I	Backup supply voltage
37	GND	-	Ground
38	Reserved	-	Connect to Ground
39	Reserved	-	Leave open
40	Reserved	-	Leave open
41	GND	-	Ground
42	TXD / SPI_SDO	O	UART1 output if D_SEL = 1 (or open). SPI_SDO if D_SEL = 0
43	RXD / SPI_SDI	I	UART1 input if D_SEL = 1 (or open). SPI_SDI if D_SEL = 0
44	SDA / SPI_CS_N	I/O	I2C Data if D_SEL = 1 (or open). SPI Chip Select if D_SEL = 0
45	SCL / SPI_CLK	I/O	I2C Clock if D_SEL = 1 (or open). SPI Clock if D_SEL = 0
46	TX_READY	O	TX_Buffer full and ready for TX of data
47	D_SEL	I	Interface select for pins 42-45
48	GND	-	Ground
49	RESET_N	I	RESET_N
50	SAFEBOOT_N	I	SAFEBOOT_N (for future service, updates and reconfiguration, leave OPEN)
51	EXTINT	I	External interrupt pin

Pin no.	Name	I/O	Description
52	Reserved	-	Reserved
53	TIMEPULSE	O	Time pulse
54	Reserved	-	Reserved

Table 15: ZED-X20P pin assignment

3.2 Pin state

Table 16 defines the state of the PIOs and **RESET_N** pins in different modes. The functions of the PIOs are as defined in the default configuration.

Pin no.	Default function	Continuous mode	Software standby mode	Safe boot mode
47	D_SEL = open	Input pull-up	Input pull-up	Input pull-up
	D_SEL = GND	High-Z	Input pull-down	High-Z
43	RXD	Input pull-up	Input pull-up	Input pull-up
	SPI_SDO	High-Z	Input pull-up	Input pull-up
42	TXD	Output	Input pull-up	Output
	SPI_SDI	Output ¹⁴	Input pull-up	Output ¹⁴
44	SDA	Input pull-up / Output	Input pull-up	Input pull-up / Output
	SPI_CS_N	High-Z	High-Z	High-Z
45	SCL	Input pull-up	Input pull-up	Input pull-up
	SPI_SLK	High-Z	High-Z	High-Z
53	TIMEPULSE	Output	Input pull-up	Output low
50	SAFEBOOT_N	Input pull-up	Input pull-up	Input pull-up
51	EXTINT	Input pull-up	Input pull-up	Input pull-up
26	RXD2	Input pull-up	Input pull-up	Input pull-up
27	TXD2	Output	Input pull-up	Output
49	RESET_N	Input pull-up	Input pull-up	Input pull-up

Table 16: Pins state


In reset mode (**RESET_N** = low), all PIOs are configured as input pull-up.



In hardware backup mode (**VCC** = 0 V), PIOs must not be driven.

¹⁴ If SPI CS = low. Otherwise it is configured as an input pull-up.

4 Electrical specifications

4.1 Absolute maximum ratings

-  CAUTION. Risk of device damage. Exceeding the absolute maximum ratings may affect the lifetime and reliability of the device or permanently damage it. Do not exceed the absolute maximum ratings.
-  CAUTION. This product is not protected against overvoltage or reversed voltages. Use appropriate protection to avoid device damage from voltage spikes exceeding the specified boundaries.

Parameter	Symbol	Condition	Min	Max	Units
Power supply voltage	VCC		-0.5	3.6	V
Voltage ramp on VCC ¹⁵			20	8000	µs/V
Backup battery voltage	V_BCKP		-0.5	3.6	V
Voltage ramp on V_BCKP ¹⁵			20		µs/V
Input pin voltage	Vin	VCC ≤ 3.1 V	-0.5	VCC + 0.5	V
		VCC > 3.1 V	-0.5	3.6	V
VCC_RF output current	ICC_RF			300	mA
Input power at RF_IN	Prfin	Source impedance = 50 Ω, continuous wave		10	dBm
Storage temperature	Tstg		-40	+85	°C

Table 17: Absolute maximum ratings

4.2 Operating conditions

-  Extreme operating temperatures can significantly impact the specified values. If an application operates near the min or max temperature limits, ensure the specified values are not exceeded.

Parameter	Symbol	Condition	Min	Typical	Max	Units
Power supply voltage	VCC		2.7	3.0	3.6	V
Backup battery voltage	V_BCKP		1.65		3.6	V
Backup battery current ^{16, 17}	I_BCKP			32		µA
SW backup current	I_SWBCKP			93		µA
Input pin voltage range	Vin		0		VCC	V
Digital IO pin low level input voltage	Vil				0.4	V
Digital IO pin high level input voltage	Vih		0.8 * VCC			V
Digital IO pin low level output voltage	Vol	Iol = 2 mA ¹⁸			0.4	V
Digital IO pin high level output voltage	Voh	Ioh = 2 mA ¹⁸	VCC - 0.4			V
DC current through any digital I/O pin (except supplies)	Ipin				5	mA
Pull-down resistors on GPIOs	Rpd		34	62	142	kΩ

¹⁵ Exceeding the ramp speed may permanently damage the device

¹⁶ To measure the I_BCKP the receiver should first be switched on, i.e. VCC and V_BCKP is available. Then set VCC to 0 V while the V_BCKP remains available. Afterward measure the current consumption at the V_BCKP.

¹⁷ The value has been characterized at 25 °C ambient temperature.

¹⁸ TIMEPULSE has 4 mA current drive/sink capability

Parameter	Symbol	Condition	Min	Typical	Max	Units
Pull-up resistance for SCL, SDA, EXTINT	R _{pu}		8	16	40	kΩ
Pull-up resistance for D_SEL, RXD, TXD, SAFEBOOT_N	R _{pu}		35	67	205	kΩ
Pull-up resistance for RESET_N	R _{pu}		7	10	13	kΩ
VCC_RF voltage	VCC_RF			VCC – 0.1		V
VCC_RF output current	ICC_RF				50	mA
Input impedance at RF_IN	Z _{in}			50		Ω
Receiver chain noise figure (L1\B1\E1)	NF _{tot}			6.5		dB
Receiver chain noise figure (L2)	NF _{tot}			6		dB
Receiver chain noise figure (L5\E5a\B2a)	NF _{tot}			5		dB
External gain (at RF_IN)	Ext_gain	Normal gain mode (default)	17		40	dB
External gain (at RF_IN)	Ext_gain	Low gain mode	20		50	dB
Operating temperature	Topr		–40	+25	+85	°C

Table 18: Operating conditions

 If L-band functionality is required, contact u-blox.

4.3 Indicative power requirements

Table 19 provides examples of typical current requirements. The given values are total system supply current for a possible application including RF and baseband sections. The values have been measured at 25 °C ambient temperature.

 The actual power requirements vary depending on the firmware (FW) version used, external circuitry, number of satellites tracked, signal strength, type and time of start, duration, and conditions of test.

Symbol	Parameter	Conditions	GPS+GAL+BDS	GPS	Unit
I _{PEAK}	Peak current	Acquisition	80	70	mA
I _{VCC}	VCC current	Acquisition	68	55	mA
I _{VCC}	VCC current	Tracking	64	55	mA

Table 19: Currents to calculate the indicative power requirements

5 Communications interfaces

The ZED-X20P receiver supports communication over UART, SPI and I2C.

All the inputs have internal pull-up resistors in normal operation and can be left open if not used. All the PIOs are supplied by V_{IO}, therefore all the voltage levels of the PIO pins are related to V_{IO} supply voltage.

5.1 UART

The UART interfaces support configurable baud rates. Hardware flow control is not supported.

The UART1 is enabled if **D_SEL** pin of the module is left open or "high".

Symbol	Parameter	Min	Max	Unit
R _U	Baud rate	4800	8000000	bit/s
Δ _{Tx}	Tx baud rate accuracy	-1%	+1%	-
Δ _{Rx}	Rx baud rate tolerance	-2.5%	+2.5%	-

Table 20: ZED-X20P UART specifications

5.2 SPI

The SPI interface is disabled by default. The SPI interface shares pins with UART1 and I2C and can be selected by setting **D_SEL** = 0. The SPI interface can be operated in peripheral mode only. The SPI transfer rate based on load capacitance is shown in [Table 21](#).

Load capacitance (pF)	Min transfer rate (kB/s)	Max transfer rate (kB/s)	Max clock frequency (MHz)
2	880	950	12.80
20	770	920	10.20
60	620	880	7.25

Table 21: SPI transfer rate based on load capacitance for ZED-X20P

The SPI timing parameters for peripheral operation are defined in [Figure 3](#). Default SPI configuration is CPOL = 0 and CPHA = 0.

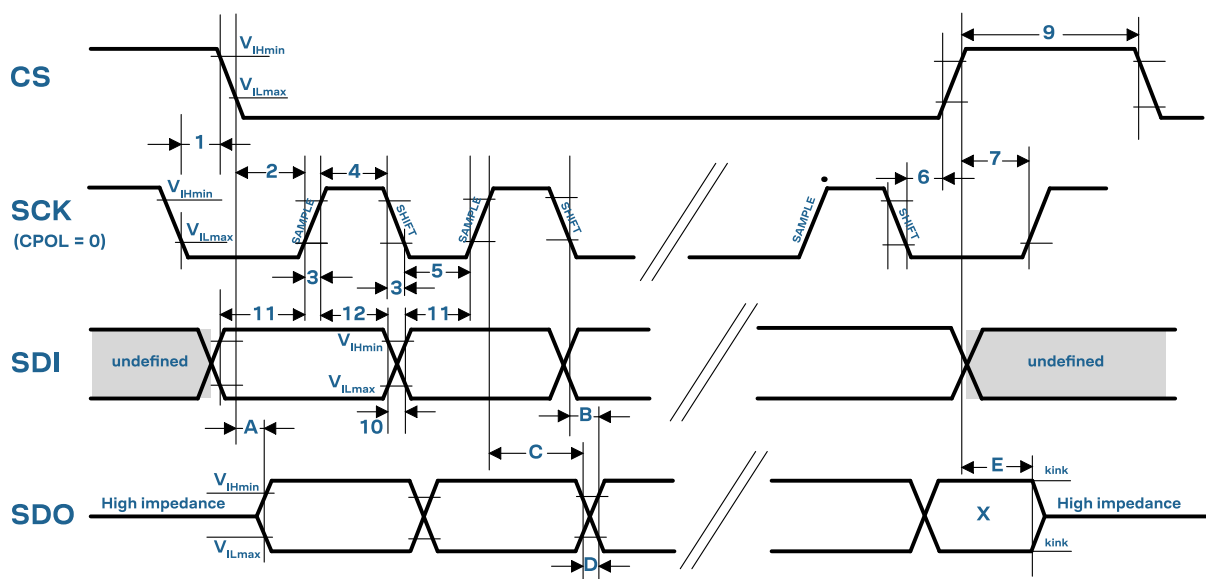


Figure 3: ZED-X20P SPI specification mode 1: CPHA=0 SCK = 5.33 MHz

Symbol	Parameter	Min	Max	Unit
1	CS deassertion hold time	8	-	ns
2	Chip select time (CS to SCK)	11	-	ns
3	SCK rise/fall time	-	5	ns
4	SCK high time	39	-	ns
5	SCK low time	39	-	ns
6	Chip deselect time (SCK falling to CS)	3	-	ns
7	Chip deselect time (CS to SCK)	510	-	ns
9	CS high time	511	-	ns
10	SDI transition time	-	5	ns
11	SDI setup time	6	-	ns
12	SDI hold time	3	-	ns

Table 22: SPI peripheral input timing parameters 1 - 12

Symbol	Parameter	Min	Max	Unit
A	SDO data valid time (CS)	15	31	ns
B	SDO data valid time (SCK), weak driver mode	15	29	ns
C	SDO data hold time	49	68	ns
D	SDO rise/fall time, weak driver mode	3	8	ns
E	SDO data disable lag time	7	20	ns

Table 23: SPI peripheral timing parameters A - E, 2 pF load capacitance

Symbol	Parameter	Min	Max	Unit
A	SDO data valid time (CS)	17	42	ns
B	SDO data valid time (SCK), weak driver mode	16	39	ns
C	SDO data hold time	52	88	ns
D	SDO rise/fall time, weak driver mode	2	21	ns
E	SDO data disable lag time	7	20	ns

Table 24: SPI peripheral timing parameters A - E, 20 pF load capacitance

Symbol	Parameter	Min	Max	Unit
A	SDO data valid time (CS)	20	62	ns
B	SDO data valid time (SCK), weak driver mode	19	59	ns
C	SDO data hold time	58	128	ns
D	SDO rise/fall time, weak driver mode	7	38	ns
E	SDO data disable lag time	7	20	ns

Table 25: SPI peripheral timing parameters A - E, 60 pF load capacitance

5.3 I2C

An I2C interface is available for communication with an external host CPU in I2C standard mode, fast mode and fast mode plus. The interface can be operated only in peripheral mode with the maximum bit rate of 1000 kbit/s. The interface can make use of clock stretching by holding the SCL line LOW to pause a transaction. In this case, the bit transfer rate is reduced. The maximum clock stretching time is 20 ms. The I2C timing parameters are defined in Figure 4 and specified in Table 26.

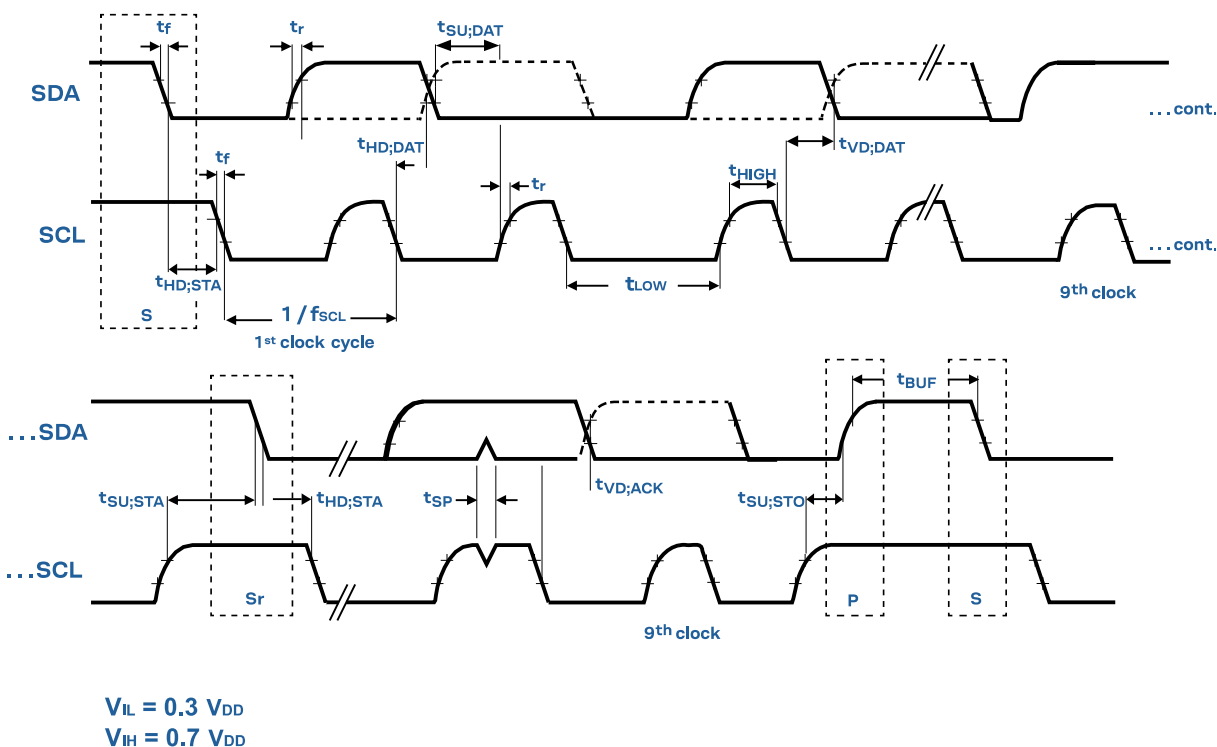


Figure 4: ZED-X20P I2C peripheral specification

Symbol	Parameter	Standard mode		Fast mode		Fast mode plus		Unit
		Min	Max	Min	Max	Min	Max	
f_{SCL}	SCL clock frequency	-	100	-	400	-	1000	kHz
$t_{HD;STA}$	Hold time (repeated) START condition	4.0	-	0.6	-	0.26	-	μs
t_{LOW}	Low period of the SCL clock	4.7	-	1.3	-	0.26	-	μs
t_{HIGH}	High period of the SCL clock	4.0	-	0.6	-	0.5	-	μs
$t_{SU;STA}$	Setup time for a repeated START condition	4.7	-	0.6	-	0.26	-	μs

Symbol	Parameter	Standard mode		Fast mode		Fast mode plus		Unit
		Min	Max	Min	Max	Min	Max	
$t_{HD;DAT}^{19, 20}$	Data hold time	0	-	0	-	0	-	μs
$t_{SU;DAT}$	Data setup time	250	-	100	-	50	-	ns
t_r	Rise time of both SDA and SCL signals	-	1000	-	300	-	120	ns
t_f	Fall time of both SDA and SCL signals	-	300	-	300	-	120	ns
$t_{SU;STO}$	Setup time for STOP condition	4.0	-	0.6	-	0.26	-	μs
t_{BUF}	Bus-free time between a STOP and START condition	4.7	-	1.3	-	0.5	-	μs
$t_{VD;DAT}^{20}$	Data valid time	-	3.45	-	0.9	-	0.45	μs
$t_{VD;ACK}^{20}$	Data valid acknowledge time	-	3.45	-	0.9	-	0.45	μs
V_{nL}	Noise margin at the low level	0.1 VCC	-	0.1 VCC	-	0.1 VCC	-	V
V_{nH}	Noise margin at the high level	0.2 VCC	-	0.2 VCC	-	0.2 VCC	-	V
C_b	Capacitive load for each bus line	-	400	-	400	-	550	pF

Table 26: ZED-X20P I2C peripheral timings and specifications


The I2C interface is only available with the UART default mode. If the SPI interface is selected by using **D_SEL** = 0, the I2C interface is not available.

5.4 Default interface settings

Interface	Settings
UART1 output	38400 baud, 8 bits, no parity bit, 1 stop bit. NMEA protocol with GGA, GLL, GSA, GSV, RMC, VTG, TXT messages are output by default. UBX and RTCM 3.4 protocols are enabled by default but no output messages are enabled by default.
UART1 input	38400 baud, 8 bits, no parity bit, 1 stop bit. UBX, NMEA and RTCM 3.4 input protocols are enabled by default.
UART2 output	38400 baud, 8 bits, no parity bit, 1 stop bit. RTCM 3.4 protocol is enabled by default but no output messages are enabled by default. NMEA protocol is disabled by default.
UART2 input	38400 baud, 8 bits, no parity bit, 1 stop bit. RTCM 3.4 protocol is enabled by default. SPARTN protocol is enabled by default. NMEA protocol is disabled by default.
I2C	Available for communication in the Fast-mode with an external host CPU in peripheral mode only. Default messages activated as in UART1. Input/output protocols available as in UART1. Maximum bit rate 400 kb/s.
SPI	Allow communication to a host CPU, operated in peripheral mode only. Default messages activated as in UART1. Input/output protocols available as in UART1. SPI is not available unless D_SEL pin is set to low (see section D_SEL interface in Integration manual[1]).

Table 27: Default interface settings

¹⁹ External device must provide a hold time of at least one transition time (max 300 ns) for the SDA signal (with respect to the min V_{IH} of the SCL signal) to bridge the undefined region of the falling edge of SCL.

²⁰ The maximum $t_{HD;DAT}$ must be less than the maximum $t_{VD;DAT}$ or $t_{VD;ACK}$ with a maximum of 0.9 μs by a transition time. The maximum must only be met if the device does not stretch the low period t_{LOW} of the SCL signal. If the clock stretches the SCL, the data must be valid by the set-up time before it releases the clock.



Refer to the applicable Interface description for information about further settings.



By default, ZED-X20P outputs NMEA messages that include satellite data for all GNSS bands being received. This results in a high NMEA output load for each navigation period. Make sure the UART baud rate used is sufficient for the selected navigation rate and the number of GNSS signals being received.

6 Mechanical specifications

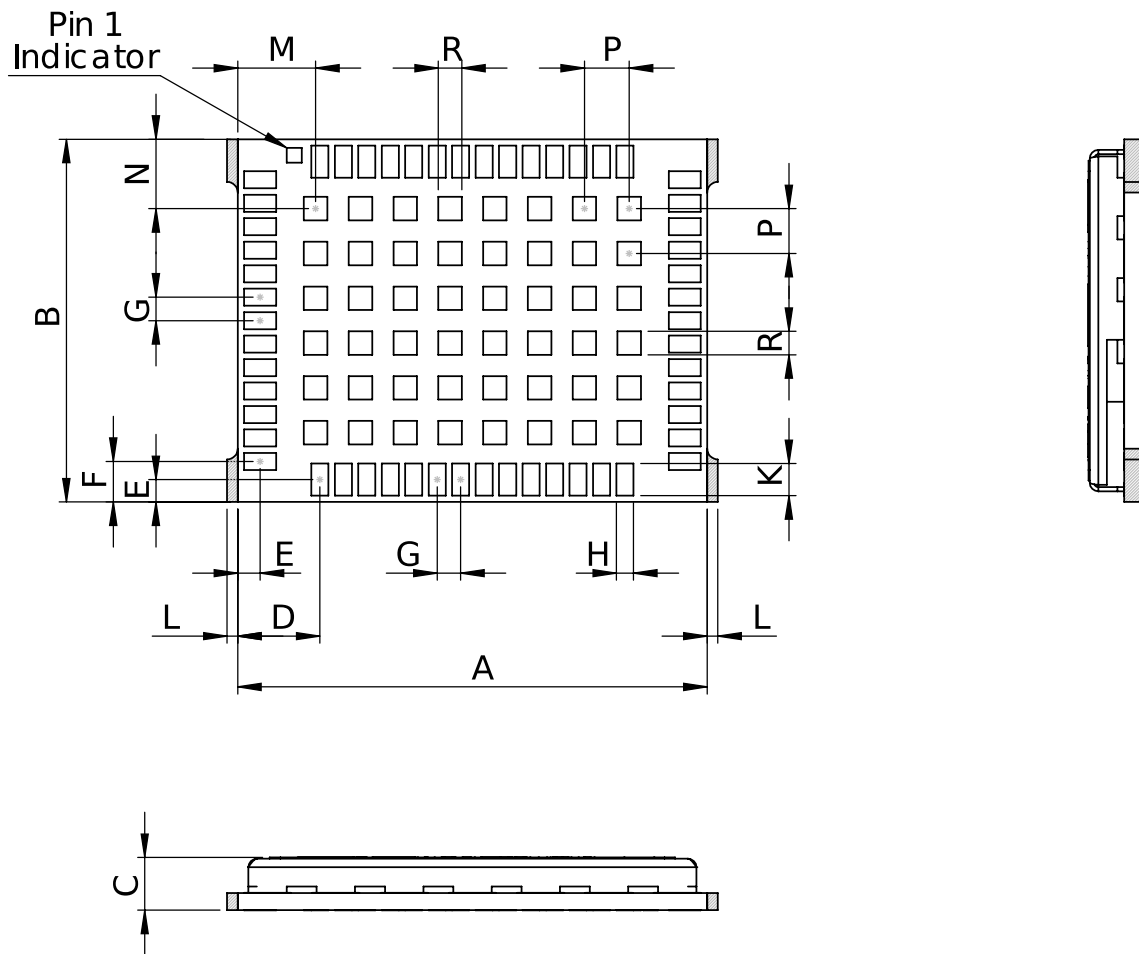


Figure 5: ZED-X20P mechanical drawing

Symbol	Min (mm)	Typical (mm)	Max (mm)
A	21.80	22.00	22.20
B	16.80	17.00	17.20
C	2.20	2.40	2.60
D	3.65	3.85	4.05
E	0.85	1.05	1.25
F	1.70	1.90	2.10
G	1.05	1.10	1.15
H	0.70	0.80	0.96
K	1.20	1.50	1.80
M	3.45	3.65	3.85
N	3.05	3.25	3.45
P	2.05	2.10	2.15

Symbol	Min (mm)	Typical (mm)	Max (mm)
R	0.88	1.10	1.32
L	0.00		0.30
Weight		1.6 g	

Table 28: ZED-X20P mechanical dimensions


The mechanical picture of the de-paneling residual tabs (L) is an approximate representation. The shape and position may vary, but the overall size of the residual tabs remains within the maximum dimensions even when the tab sizes differ.



Take the size of the de-paneling residual tabs into account when designing the component keep-out area.

7 Qualifications and approvals

Type	Description
Quality and reliability	
Product qualification	Qualified according to u-blox qualification policy, based on a subset of AEC-Q104.
Chip qualification	Modules are based on AEC-Q100 qualified GNSS chips.
Manufacturing	Manufactured at ISO/TS 16949 certified sites.
Environmental	
RoHS compliance	Yes
Moisture sensitivity level (MSL) ^{21, 22}	4
Type approvals	
European RED certification (CE)	Yes
UK conformity assessment (UKCA)	Yes

Table 29: Qualifications and approvals

²¹ For the MSL standard, see IPC/JEDEC J-STD-020 and J-STD-033 [3].

²² For more information regarding moisture sensitivity levels, labeling, storage, and drying, see the Product packaging reference guide [4].

8 Packaging

The ZED-X20P modules are delivered as hermetically sealed, reeled tapes in order to enable efficient production, production lot set-up and tear-down. For more information about packaging, see the Product packaging reference guide [4].

8.1 Reels

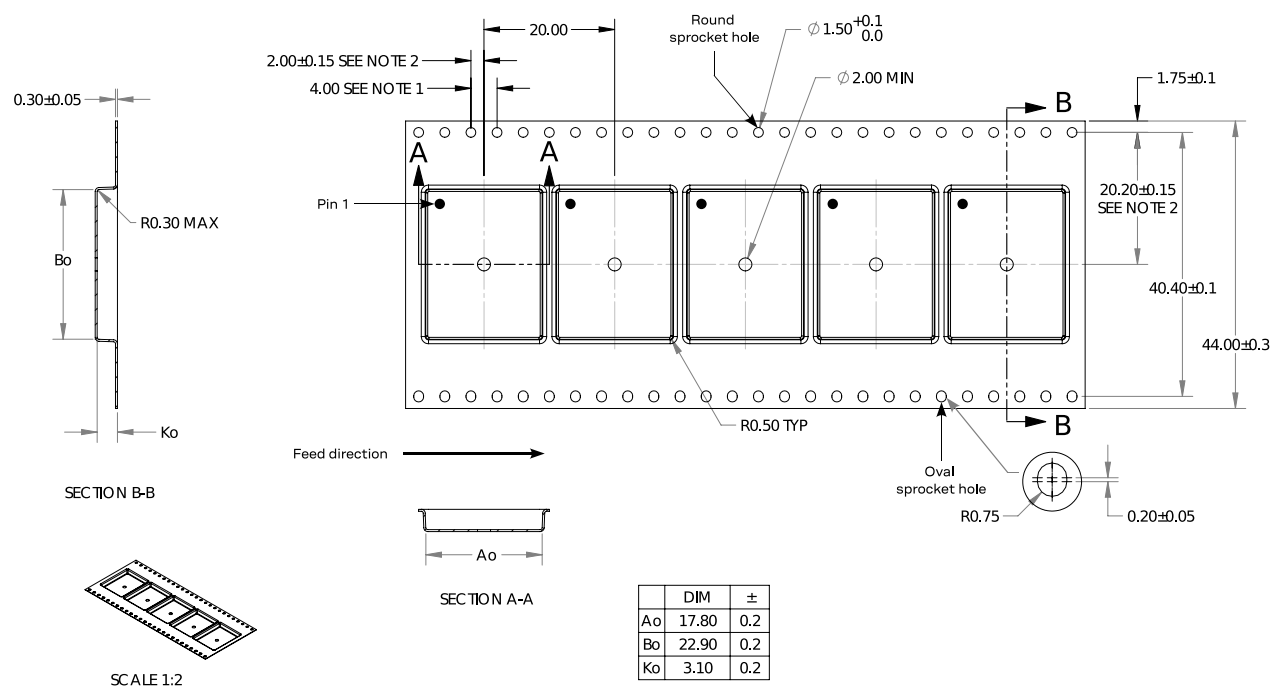
The ZED-X20P modules are deliverable in quantities of 250 pieces on a reel. They are shipped on reel type B, as specified in the Product packaging reference guide [4].

Package	Reel type	Delivery quantity
SMD	B2	250

Table 30: Reel information for modules

8.2 Tapes

Figure 6 illustrates the feed direction, component orientation, and tape dimensions (mm). The tape feeds into the pick-and-place machine from the reel on the left side of the figure and moves to the right.



- NOTES:
1. 10 SPROCKET HOLE PITCH CUMULATIVE TOLERANCE ± 0.2
 2. POCKET POSITION RELATIVE TO SPROCKET HOLE MEASURED AS TRUE POSITION OF POCKET, NOT POCKET HOLE.
 3. Ao AND Bo ARE MEASURED ON A PLANE AT A DISTANCE "R" ABOVE THE BOTTOM OF THE POCKET.

Figure 6: ZED-X20P Tape dimensions (mm)

9 Soldering

For information on reflow soldering, see IPC/JEDEC J-STD-020 [\[3\]](#) and Integration manual[\[1\]](#).

10 Product marking and ordering information

This section provides information about product marking and ordering.

10.1 Product marking

The product marking provides information on ZED-X20P and its revision, as in [Figure 7](#). For a description of the product marking, see [Table 31](#)

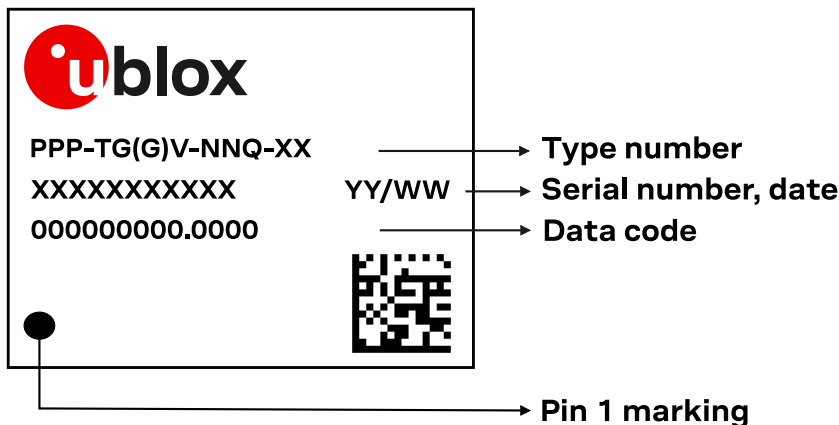


Figure 7: Example of ZED-X20P marking

Code	Meaning	Example
PPP	Form factor	ZED
TG(G)	Platform	X20 = u-blox X20
V	Variant	P = High precision
NN	Major product version	00, 01, ..., 99
Q	Product grade	A = Automotive B = Professional C = Standard
XX	Revision	Hardware and firmware versions
XXXXXXXXXXXX	Serial number	Alphanumeric characters, e.g. BN600001181
000000000.0000	Data code	Alphanumeric characters
YY/WW or YYWW	Production date	Year/week, e.g. 24/04 or 2404
QR code	For internal/technical use	

Table 31: Description of product marking

10.2 Product identifiers

The ZED-X20P marking features three product identifiers: product name, ordering code and type number. The product name identifies all u-blox products, independent of packaging and product grade, and it is used in documentation such as this data sheet. The ordering code includes the major product version and product grade, while the type number additionally includes the hardware and firmware versions.

[Table 32](#) describes the three different product identifiers used in the ZED-X20P product marking.

Identifier	Format	Example
Product name	PPP-TGGV	ZED-X20P

Identifier	Format	Example
Ordering code	PPP-TGGV-NNQ	ZED-X20P-00B
Type number	PPP-TGGV-NNQ-XX	ZED-X20P-00B-00

Table 32: Product identifiers

10.3 Ordering codes

Ordering code	Product	Remark
ZED-X20P-00B	ZED-X20P	Shipped with firmware HPG 2.02

Table 33: Product ordering codes

u-blox provides information on product changes affecting the form factor, size or function of the product. For the Product change notifications (PCNs), see our website at: <https://www.u-blox.com/en/product-resources>.

Related documents

- [1] Integration manual, [UBXDOC-963802114-12901](#)
- [2] Interface description, [UBXDOC-304424225-19888](#)
- [3] MSL standard IPC/JEDEC J-STD-020, www.jedec.org
- [4] Product packaging reference guide [UBX-14001652](#)

For product change notifications and regular updates of u-blox documentation, register on our website, <https://www.u-blox.com>.

Revision history

Revision	Date	Comments
R01	06-Sep-2024	Initial revision.
R02	04-Mar-2025	Update on the supported GNSS constellations, communications interfaces, qualifications and approvals, and packaging sections. General text enhancement.
R03	23-May-2025	Updated sections: • Performance • Firmware features • Pin assignment • Operating conditions • Indicative power requirements • Communications interfaces
R04	22-Sept-2025	Updated sections: • Firmware version • Performance • Operating conditions • Qualifications and approvals

Contact

u-blox AG

Address: Zürcherstrasse 68
8800 Thalwil
Switzerland

For further support and contact information, visit us at www.u-blox.com/support.